



University of
Nottingham
UK | CHINA | MALAYSIA

COMP3055

Machine Learning

Topic 1 – Introduction

2025 Autumn

COMP3055 Machine Learning

Ying WENG

Teaching for 6 weeks

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Zheng LU

Teaching for 6 weeks (Module Convenor)

Office: PMB-426

Office hour: Monday 14:00 – 16:00

Email: zheng.lu@nottingham.edu.cn

- Lecture
 - Monday 16:00-18:00, YFB-226
 - Wednesday 12:00-13:00, IAMET-326
- Lab
 - Thursday 11:00-13:00, IAMET-406
 - For the 1st Teaching Week, there is no lab on Thursdat 25th Sep

Lecture Schedule

- Plan ahead
- All material on Moodle
- Regular update

Lab Schedule

- Lab sessions will provide you hands on experiences what you learned during the lecture sessions
- Lab materials will be published on Moodle

Summary of Content

- Provide you with an introduction to **machine learning**, **pattern recognition**, and **data mining** techniques
- Enable you to consider both systems which are able to develop their own rules from trial-and-error experience to solve problems
- Find patterns in data without any supervision. Data mining techniques will make generation of new knowledge possible, including big datasets. This is now fashionably termed 'Big Data' science
- Cover a range of topics including: machine learning foundations; pattern recognition foundations; applications of machine learning; data mining techniques and evaluating hypotheses; artificial neural networks; deep learning
- You'll spend **six hours** each week (on average) **in lectures and computer practice** for this module

Assessments

❖ Exam: 70%

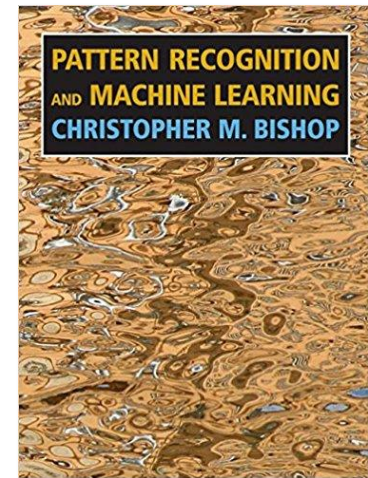
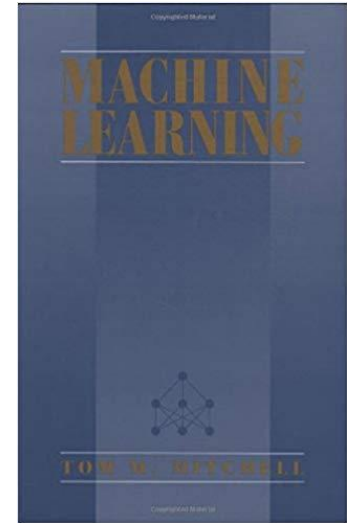
- 2 hour written examination

❖ Coursework: 30%

- 1 piece of individual programming assignment

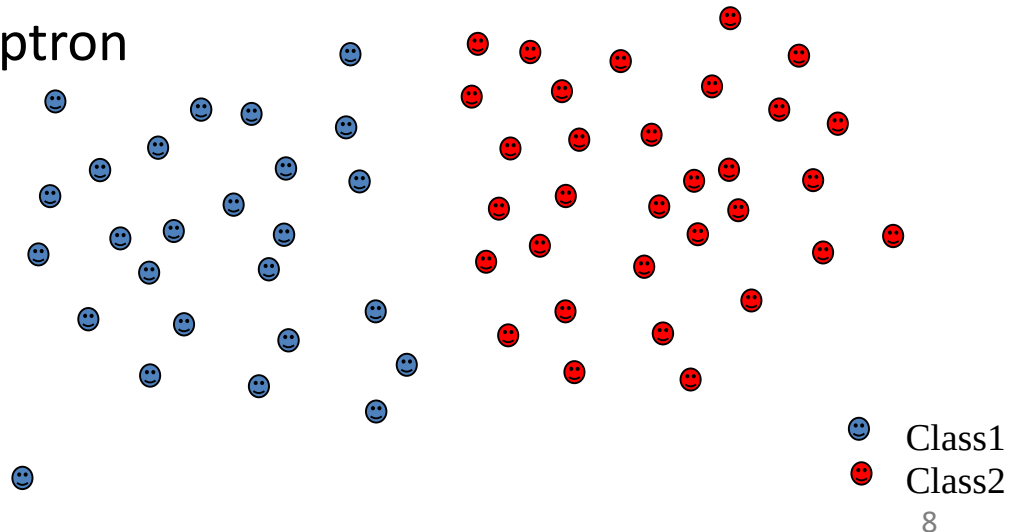
Textbooks

- Tom M. Mitchell, **Machine Learning**, McGraw-Hill
- Chris Bishop, **Pattern Recognition and Machine Learning**, Springer
- Some of recent technical papers



Topics

- Design learning systems
- Data collection
- Learning theory practice
- Instance based learning, KNN etc.
- Bayesian learning
- Data processing representation
- Unsupervised learning, clustering, K-Means etc.
- Decision tree, random forest
- Perceptron, multilayer perceptron
- Support vector machine
- Deep learning, CNN etc.
- ...
- (to be continued)



How to Get 70+

- **Studying...**

- You are recommended to study the relevant notes before attending the lecture or lab.
- Review as soon as possible to maximize retention.



- **Practice...**

- **Do the lab exercise yourself** and repeat the practice for better learning.
- If you get help on the labs, don't just blindly accept it, but try to understand what each part of the code is doing.
- Do the Math in the lecture for better understanding.

- **Assignments...**

- Start work on the assignment **when they are released**, and come up with a good plan to finish it.
- It will take **longer** than you expect to fix problems in your program, so make sure you have plenty of time to complete.

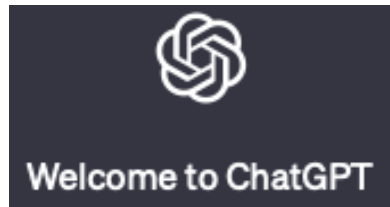
Do the thing right, Do the right thing



A Few Quotes

- “A breakthrough in machine learning would be worth ten Microsofts” (Bill Gates, Chairman, Microsoft)
- “Machine learning is the next Internet” (Tony Tether, Director, DARPA)
- Machine learning is the hot new thing” (John Hennessy, President, Stanford)
- “Machine learning is going to result in a real revolution” (Greg Papadopoulos, CTO, Sun)
- “Machine learning is today’s discontinuity” (Jerry Yang, CEO, Yahoo)

ChatGPT

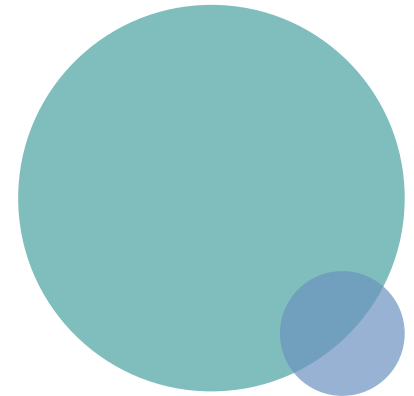


- ❖ An artificial intelligence (AI) chatbot developed by OpenAI and release in November 2022
- ❖ Built on top of GPT-4 foundational large language model (LLM)
- ❖ Fine-tuned to transfer learning

incredible, powerful, helpful

ChatGPT has rapidly immersed itself into our lives.
I think everyone has read articles related to it.

ChatGPT is a milestone in the era of large models.



What is Machine Learning?

直观理解：

某些任务（如人脸识别）难以手写规则。

通过收集输入-输出样本对，由算法自动生成能完成任务的程序。

结果程序可能包含数百万个参数（weights），但能泛化到新样本。

核心思想：让数据代替程序员写程序。

即“Automating automation”——自动化的自动化。

- It is very hard to write programs that solve problems like recognizing a face.
 - We don't know what program to write because we don't know how our brain does it.
 - Even if we had a good idea about how to do it, the program might be horrendously complicated.
- Instead of writing a program by hand, we collect lots of examples that specify the correct output for a given input.
- A machine learning algorithm then takes these examples and produces a program that does the job.
 - The program produced by the learning algorithm may look very different from a typical hand-written program. It may contain millions of numbers.
 - If we do it right, the program works for new cases as well as the ones we trained it on.

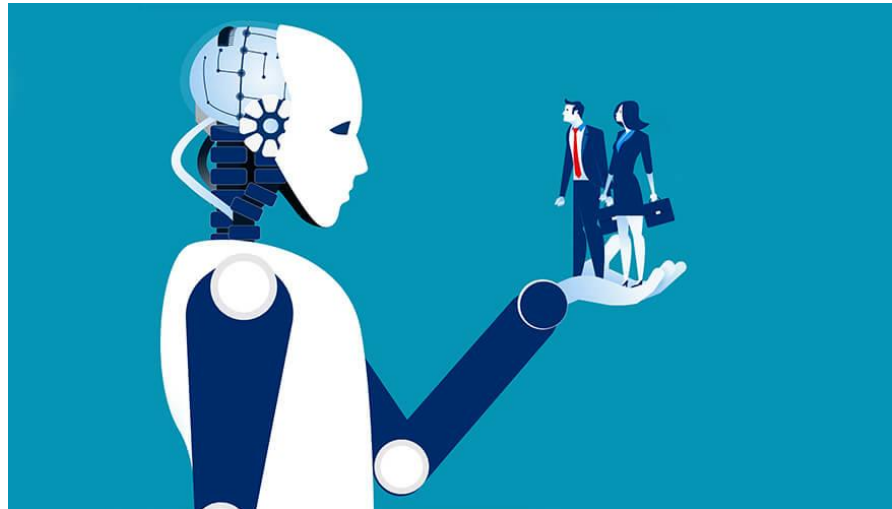
What is Machine Learning?

- Automating automation
- Getting computers to program themselves
- Writing software is the bottleneck
- Let the data do the work instead!

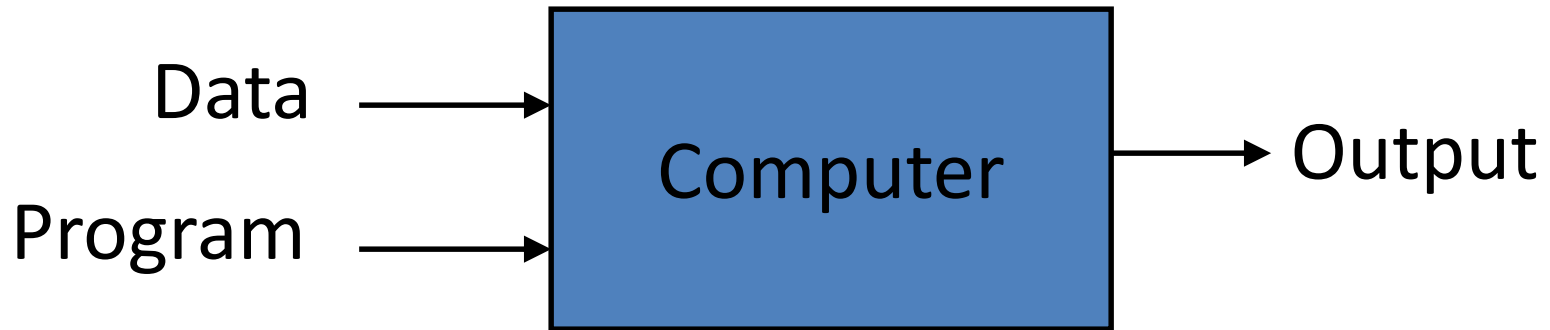


What is Machine Learning?

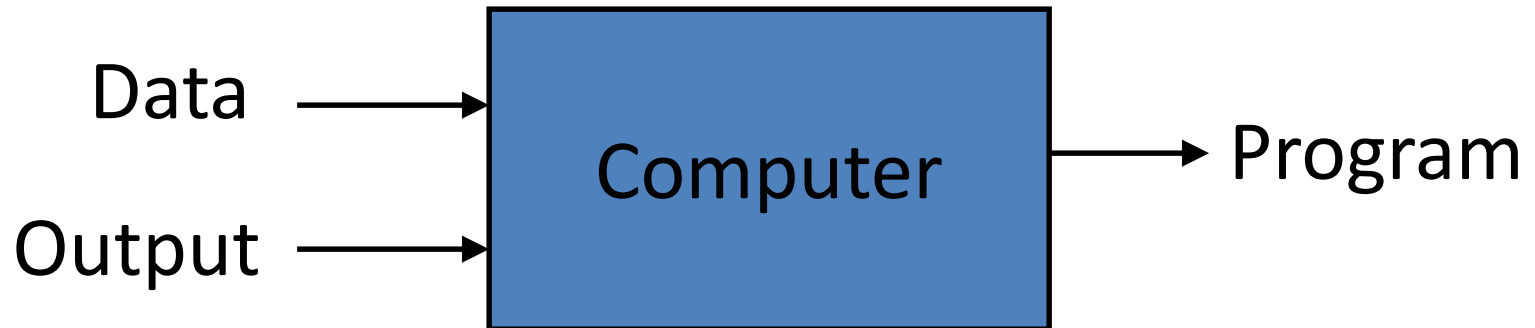
- Machine Learning is making great strides
 - Large, good data sets
 - Computer power
 - Progress in algorithms
- Many interesting applications
 - Commercial
 - Scientific



Traditional Programming



Machine Learning



Magic?

No, more like gardening

- **Seeds** = Algorithms
- 营养 **Nutrients** = Data
- **Gardener** = You
- **Plants** = Programs



Task: Handwriting Recognition

It is very hard to say what makes a 2

0 0 0 1 1 1 1 1 1 2

2 2 2 2 2 2 2 3 3 3

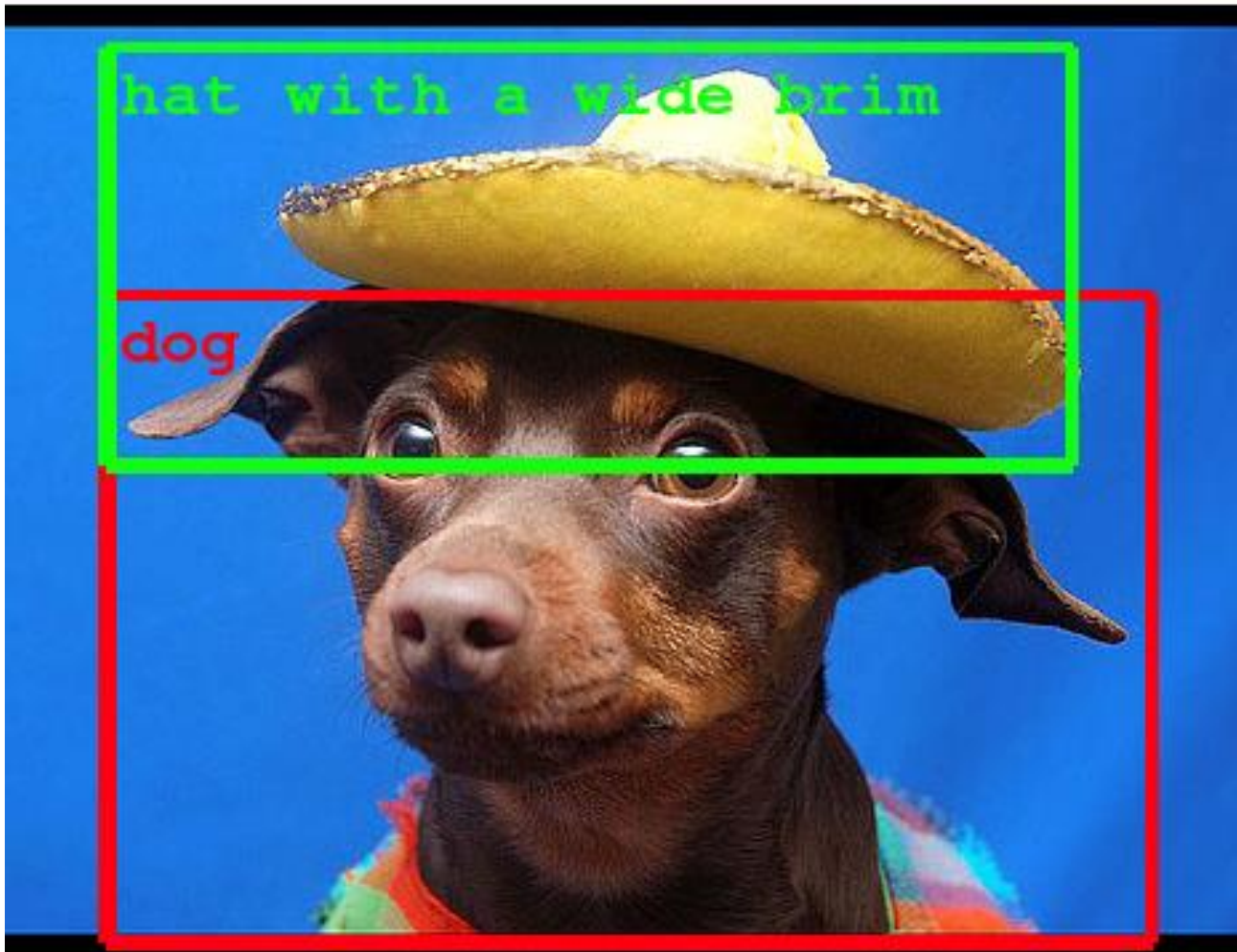
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6 6 7 7 7 7 7 8 8 8

9 9 9 9 9 9 9 9 9

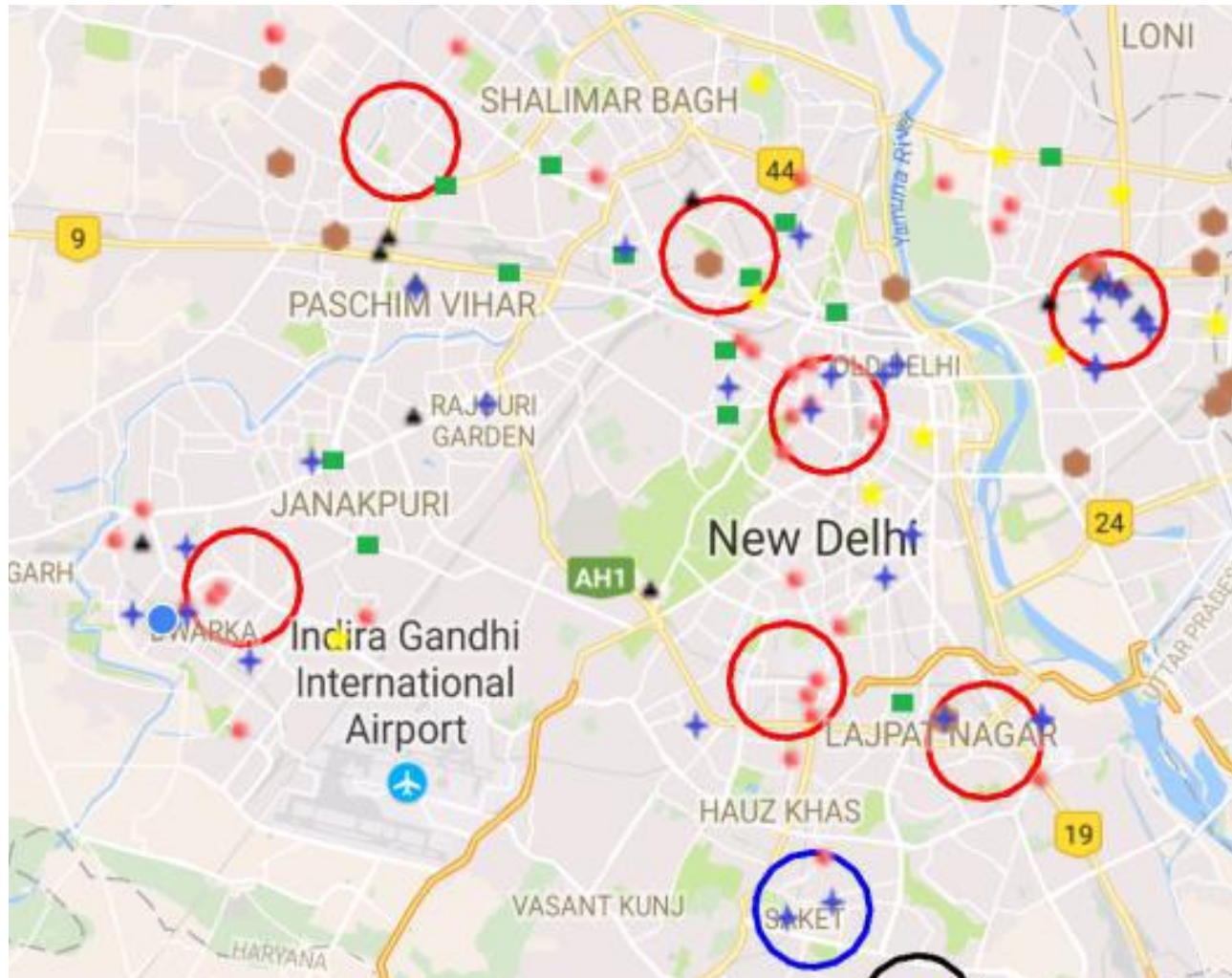
Task: Classification

Object recognition



Task: Clustering 聚类

Crime prediction using k-means clustering



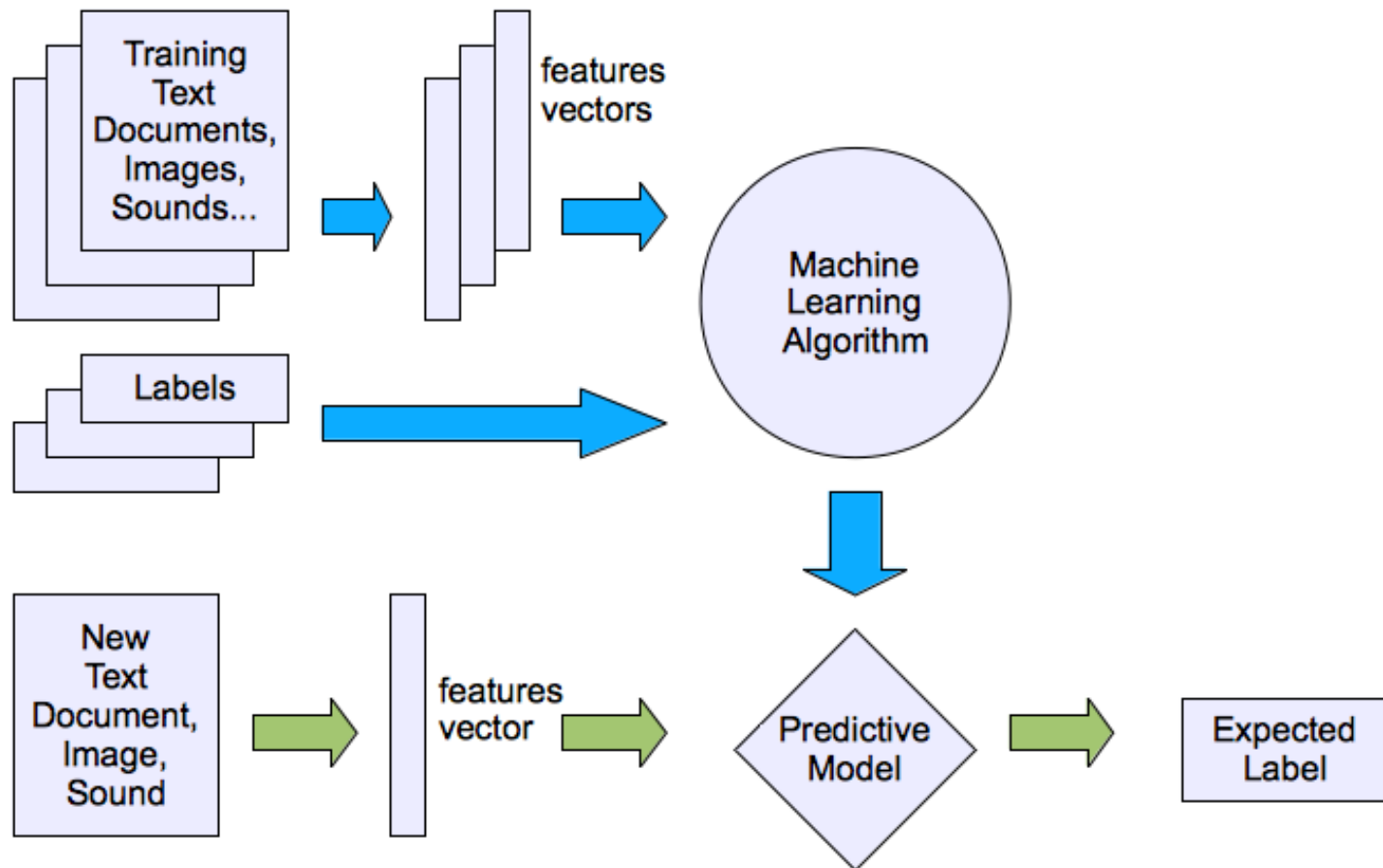
More Sample Applications

- Web search
- Healthcare
- Robotics
- Information extraction
- Finance
- E-commerce
- Computational biology
- Space exploration
- Social networks
- [Your favorite area]

网络搜索 (Web Search)
医疗健康 (Healthcare)
机器人 (Robotics)
信息抽取 (Information Extraction)
金融与电商 (Finance & E-commerce)
生物计算 (Computational Biology)
太空探索 (Space Exploration)
社交网络 (Social Networks)
等等 (甚至包括你的兴趣领域)

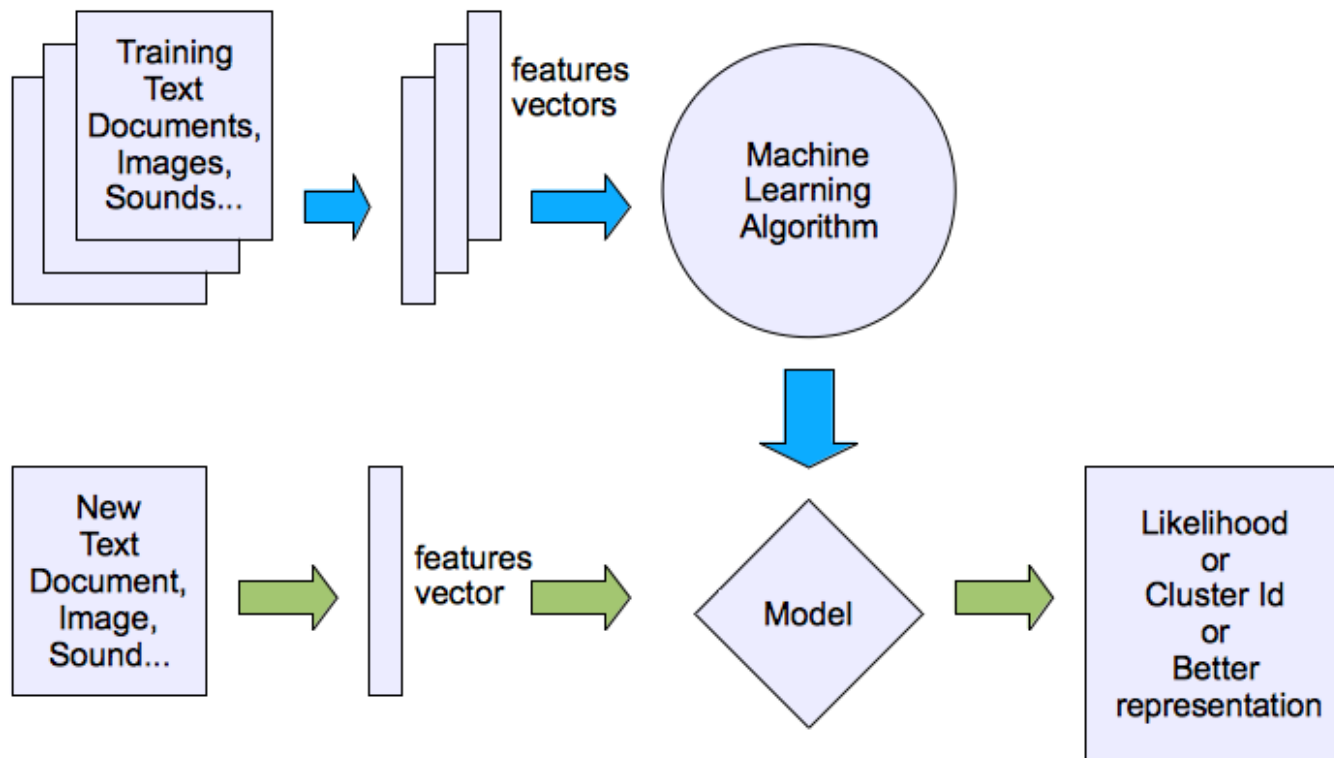
Types of Learning

- Supervised learning 监督学习：数据有标签（输入-输出对）



Types of Learning

- Unsupervised learning 无监督学习：数据无标签



Types of Learning

- **Supervised (inductive) learning**
 - Training data includes **desired outputs** 预期输出
- **Unsupervised learning**
 - Training data does **not include desired outputs**
 - This is the **new frontier** of machine learning because most big datasets do not come with **labels**
- **Semi-supervised learning**
 - Training data includes **a few desired outputs**

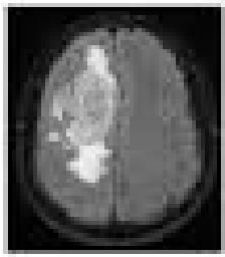
Semi-supervised Learning

Motivation: **labeled data are expensive**

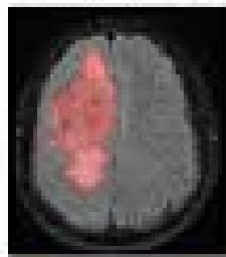
unlabeled data are free

医学领域：

- 对医学图像进行标注的过程需要专门的知识，而且往往耗时且成本高昂。



unlabeled



labeled

Medical field:

- 这是因为医学图像的复杂性、对高质量标注的需求以及训练有素的医学专业人员的稀缺。

- The process of labeling medical images requires **specialized expertise** and is often **time-consuming** and **expensive**.
- This is due to the **complexity** of medical images, the need for **high-quality annotations**, and **the limited availability of trained medical professionals**.

Framework

- **Programming languages**
 - Python
 - C++
 - ...
- **Many libraries**
 - PyTorch
 - TensorFlow
 - ...

Evaluation

- Accuracy
 - Precision and recall
 - Squared error
 - Likelihood
 - Posterior probability
 - Cost / Utility
 - Entropy
 - etc.
- Accuracy (准确率)
Precision (精确率) & Recall (召回率)
Squared Error (平方误差)
Likelihood (似然)
Posterior Probability (后验概率)
Cost / Utility (代价或效用)
Entropy (熵)
等等

Issues

- Many machine learning/AI projects fail (Gartner claims 85%)

Gartner研究表明：约85%的机器学习/AI项目失败。

原因：

提出了错误的问题 (Asking the wrong question)

解决了错误的问题 (Wrong problem)

数据不足或不合适 (Not enough/right data)

用错工具或模型 (Wrong tools/model)

缺乏衡量标准 (Wrong yardstick)

招错人 (Wrong people)

忽视伦理 (Ethics)



Reasons for Failure

- Asking the wrong question
- Trying to solve the wrong problem
- Not having enough data
- Not having the right data
- Hiring the wrong people
- Using the wrong tools
- Not having the right model
- Not having the right **yardstick**

提出错误的问题
试图解决错误的问题
数据不足
数据不正确
招聘了错误的人才
使用了错误的工具
没有合适的模型
没有合适的衡量标准



模块	核心理念
机器学习本质	用数据替代规则，自动生成程序。
学习类型	监督、无监督、半监督。
关键工具	Python、PyTorch、TensorFlow。
评估标准	准确率、召回率、代价、熵等。
应用方向	医疗、金融、机器人、搜索、社交网络等。
挑战	数据质量、伦理、解释性与泛化能力。

